Structures II
Martin Paull

Gerberettes

And how Tension and Cantilevers can open space

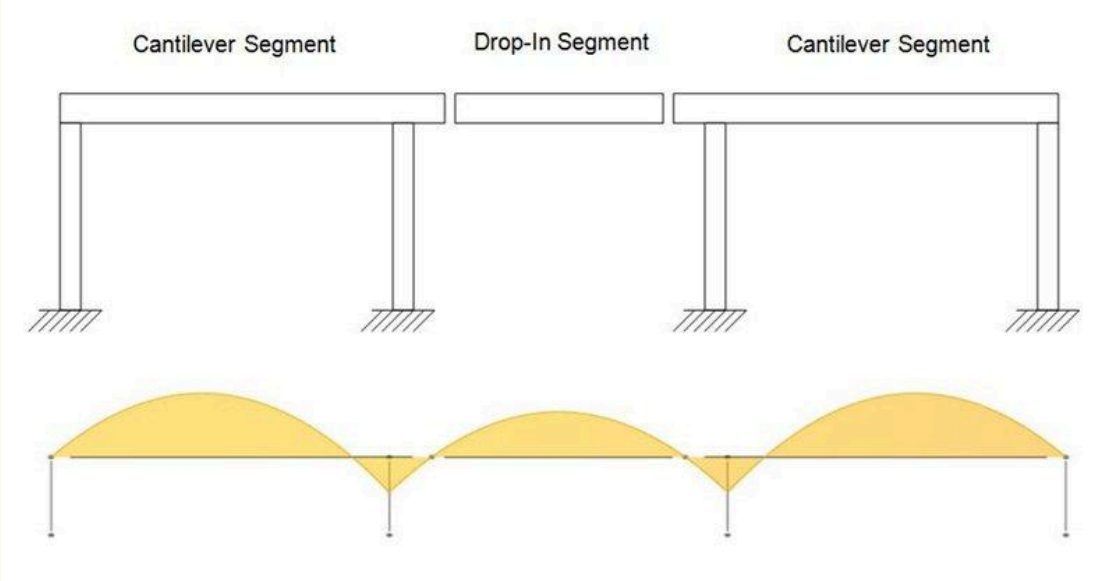
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By Alexander Morris



INTRODUCTION

The Gerberette is a system derived from the gerber beam bridge construction method, used in the construction and design of the Center Pompidou in Paris as a way to keep the floor spans clear while retaining space outside the compressive members for the circulatory pathways highlighted as the buildings facade. What interested me in this system was the fact that Tension is used as a counterbalance to the bending caused by the floors of the building (themselves supported throughout by trusses) and that the cantilevering mechanism of the gerberette allows the distance separating the bending moments acting in opposition to each other to both function as well as become so viscerally clear as a hinge mechanism dividing them around the compression member, which prioritizes at least as much acting as the rotation point for the hinge between the clockwise and counterclockwise rotations as being an anchor for the building. The efficiency of the system also fascinated me, as due to the fact that each compressive column is closer to the floors and that the floors are very slightly shorter, the presence of the cantilever provided by the gerberette and the tension members actually allows a reduction in necessary dimension to the floors and compressive columns.



Origin of Gerber Beams and counteracting cantilevered bending in construction.

Heinrich Gerber was a civil engineer working in and around the Rhine river in Germany and the towns of Bamberg and Haßfurt were where a number of his first projects in his own construction

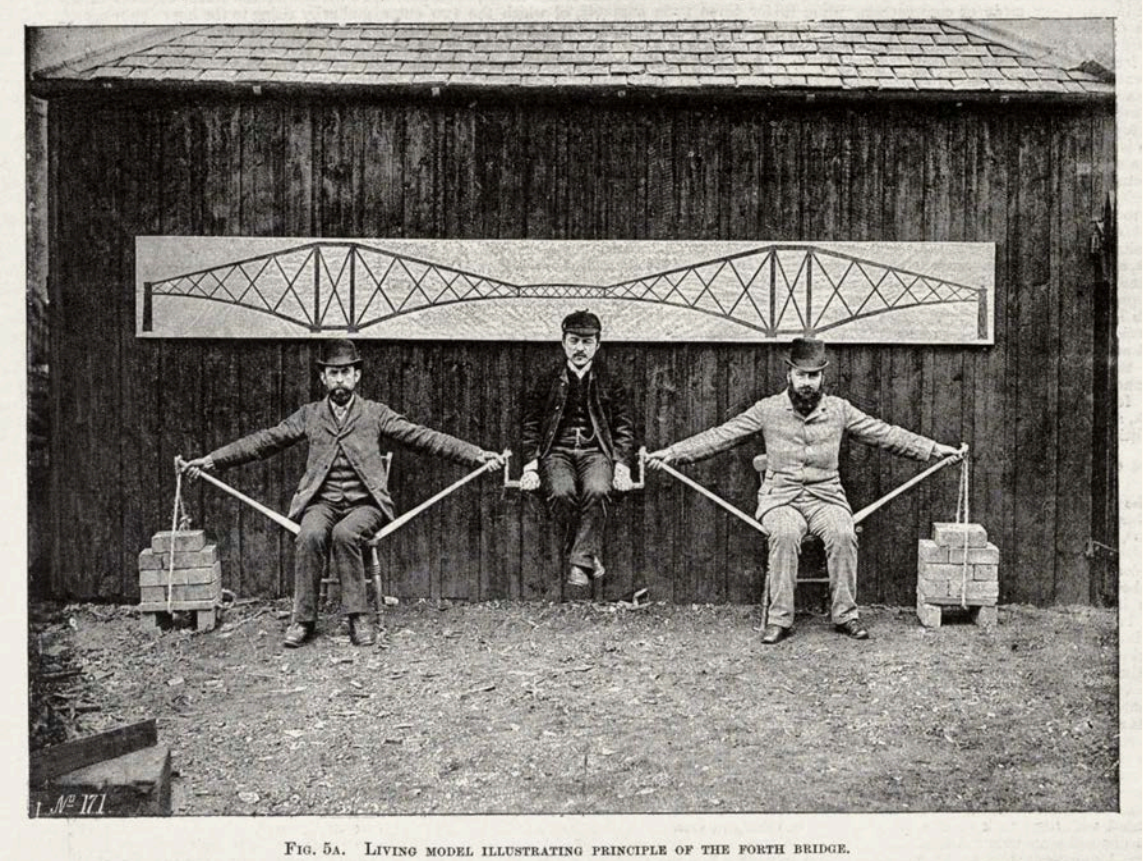


Fig. 5a. Living model illustrating principle of the Forth Bridge.

technique were realized. This method utilized cantilevered surfaces extending over compressive support structures towards the edges of the span (in this case the rivers) that connected to the shoreline for support and allowed construction to take place on the majority of the bridge without impeding existing travel and commerce. It essentially took the bending moment graph and broke it at the points of inflection to allow more modularity in construction and, eventually in the centre pompidou, in usage. The center span would then be lifted in from a boat and would cause opposite bending to the support from the shoreline, causing the structure to be in equilibrium. This method has been adapted into various other uses with time but another interesting use which still prioritizes use of an area beneath a span during construction is gerber girders used in some highway overpass construction, allowing for minimal blockage of the roadway or space underneath during phased construction, even if the center span is not raised into place in the same way.



In the 1970s, Peter Rice, Renzo Piano and Richard Rogers collaborated to build the Centre Pompidou, where they allocated the structure to form a similar set of dimensions to these span-crossing constructions, and so employed a similar system that could leverage the counteracting bending moments at small distances from each other to form a turbine blade or door handle shaped member they called the Gerberette, which rotates around the compression



column as a hinge and holds one of the building's floors at one end and has a tension member pulling it to the "shore" ground outside the compression column on the other. This spatial language they used to put circulatory paths for both the building and its inhabitants on the outside layer between the "shore" and the compression column, leaving the floor spans between them open. Since the hinge is located AT the compression columns, each part that is causing bending must cantilever away from the column to act with rotation around that point, and it brings the counteractive nature of the floors and the tension members into clear view.

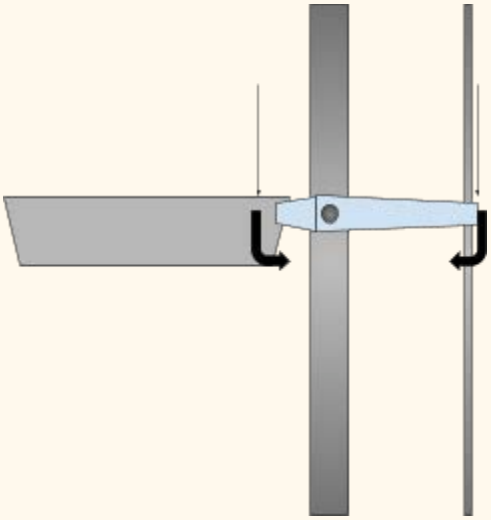
My Exhibit Project



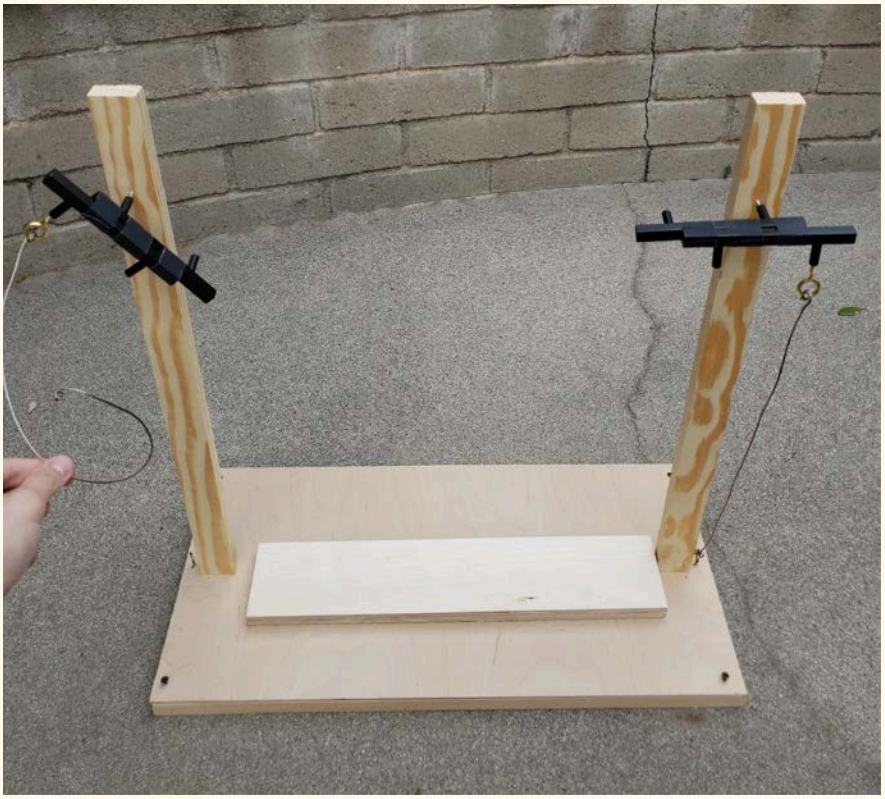


My goal with this project was to create a single floor span between two "toy"-ified gerberette members, to show how this structural system needs to find equilibrium through each side being in tension in order to hold the floor successfully. I utilized cabinet handles that, when two are taped together, allowed for cantilevering of screw holes beyond the equilibrium point for the attachment of the





tension members and a flat surface for a “floor”, or block of wood, to sit on when the system is tightened. When either tension member is taken out, the floor will fall and the system fails, but



when both are in place, the gerberettes allow for the weight of a heavy floor over the span between them, and there is a necessary space between the column and the tension members to the outside as well. In addition, one tension member can be used with a smaller floor to gradually test and see the ability of one half of the system to cantilever additional load opposite of the tension member. Ultimately, it shows the balance of the rotation between the bending forces and how the spaces between members can be left open for use.

Future Improvements and Thoughts

I really like how this came together and demonstrates the counteractive rotation of the bending moments caused by the floor and the tension members respectively and accurately downplays the role of the columns. I think there are two areas which I would improve the project as a display and exhibit beyond adding some of the images and diagrams included in this document; functional demonstrative changes and material improvements.

1. Material Improvements

I think to make this slightly more kid friendly it would be good to soften some of the materials and corners, such as the floor block being less injury-prone if it fell on someone's foot or the metal cord being sealed with rubber or something else to prevent

stabbing into fingers. The columns ended up a little taller than I would likely want to have them as well.

2. Demonstrative/Functional Improvements

As we discussed in class, it would improve the teaching capacity of the project to introduce a greater variety of attachment points (specifically AT the location of the compression member) in order for kids to be able to test them out and "see what happens". One aspect that could be improved as well is finding a better way to pin-connect the floor to the gerberette or to run the tension member through the gerberette to the floor in order to allow the floor to fail without falling completely and show where the system breaks with a little bit more specificity. Finally, adding some signage and labels as well as moment diagrams and some of the images included in this document showing how the system works around the points of inflection and works with more efficiency with the members that are there.



Conclusion

I love how clearly this mechanism displays the real rotation happening by bending moments by making the gerberette become a true hinge for the balancing of the floor and the tension and that, while the compression is important, this isn't as simple as tension vs compression yet it is as simple as clockwise vs counterclockwise. I enjoyed designing it and learning about the history and physics even more of these systems and how effortlessly this type of system can be used in all kinds of ways in architecture and design!

Resources:

Suminski, Beth, “Did You Know? RAM Steel for Gerber Beam Systems”, Bentley Colleague Blogs, Aug 2013, https://communities.bentley.com/other/old_site_member_blogs/bentley_employees/b/beth_suminski_blog/posts/did-you-know-ram-steel-for-gerber-beam-systems

“Living Model Illustrating Principle of the Forth Bridge”, Unknown photographer for Benjamin Baker. - http://www.sbe.hw.ac.uk/staff/arthur/frbpc/ForthBridgePostCards.htm, https://commons.wikimedia.org/w/index.php?curid=415051

Duncan, Adrian, "A Bridge In Haßfurt”, Fallow Media, Nov 2018, https://fallowmedia.com/2018/nov/bridge-in-hassfurt/

https://www.centrepompidou.fr/en/collection/our-building#4001